

Practical Week 7

Session 7

Task 1

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.



```
GNU nano 8.7.1 session7_memory.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

// Global variable
int global_var = 100;

// Static global variable
static int static_var = 200;

// Function represents code segment
void code_function() {
    printf("This is a function in the code segment.\n");
}

int main() {

    // Local variable - stack
    int stack_var = 300;

    // Static local variable
    static int static_local_var = 400;

    // Heap variable
    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL) {
        perror("malloc");
        return 1;
    }

    *heap_var = 500;

    printf("Linux Process Memory Address Space\n");
    printf("=====\n");

    printf("PID                : %d\n", getpid());

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

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GNU nano 8.7.1session7_memory.c *

```
printf("PID                : %d\n", getpid());

printf("\nMemory Addresses:\n");

// Code segment
printf("Code (function)      : %p\n", (void *)code_function);

// Global
printf("Global variable       : %p\n", (void *)&global_var);

// Static
printf("Static variable        : %p\n", (void *)&static_var);

// Static local
printf("Static local variable: %p\n",
      (void *)&static_local_var);

// Heap
printf("Heap variable          : %p\n", (void *)heap_var);

// Stack
printf("Stack variable         : %p\n", (void *)&stack_var);

printf("\nValues:\n");

printf("Global variable      = %d\n", global_var);
printf("Static variable     = %d\n", static_var);
printf("Static local        = %d\n", static_local_var);
printf("Heap variable       = %d\n", *heap_var);
printf("Stack variable      = %d\n", stack_var);

free(heap_var);

return 0;
}
|
```

^G Help

^X Exit

^O Write Out

^R Read File

^F Where Is

^_ Replace

^K Cut

^U Paste

^T Execute

^J Justify

```
jella_pragathi@LAPTOP-RQQGT555:~$ cd ~/OSSP
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ nano session7_memory.c
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ gcc -Wall -Wextra -g -no-pie session7_memory.c -o session7_memory
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ ./session7_memory
Linux Process Memory Address Space
=====
PID                               : 547

Memory Addresses:
Code (function)                   : 0x4011f6
Global variable                   : 0x404048
Static variable                   : 0x40404c
Static local variable             : 0x404050
Heap variable                     : 0x3a531010
Stack variable                    : 0x7ffc0486fb5c

Values:
Global variable                   = 100
Static variable                   = 200
Static local                     = 400
Heap variable                     = 500
Stack variable                    = 300
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$
```

Task 2

Make use of the /proc/maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.



```
GNU nano 8.7.1 session7_maps.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_data = 100;

int main() {

    int stack_data = 200;

    int *heap_data = malloc(sizeof(int));

    if (heap_data == NULL) {
        perror("malloc");
        return 1;
    }

    *heap_data = 300;

    printf("Memory Analysis Program\n");
    printf("=====\n");

    printf("PID: %d\n", getpid());

    printf("Global address : %p\n", (void *)&global_data);
    printf("Heap address   : %p\n", (void *)heap_data);
    printf("Stack address  : %p\n", (void *)&stack_data);

    printf("\nProcess is running...\n");
    printf("Press Ctrl+C to terminate.\n");

    while (1) {
        sleep(1);
    }

    free(heap_data);

    return 0;
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify

```
jella_pragathi@LAPTOP-RQQG555:~$ cd ~/OSSP
jella_pragathi@LAPTOP-RQQG555:~/OSSP$ nano session7_memory.c
jella_pragathi@LAPTOP-RQQG555:~/OSSP$ gcc -Wall -Wextra -g -no-pie session7_memory.c -o session7_memory
jella_pragathi@LAPTOP-RQQG555:~/OSSP$ ./session7_memory
Linux Process Memory Address Space
=====
PID : 547

Memory Addresses:
Code (function) : 0x4011f6
Global variable : 0x404048
Static variable : 0x40404c
Static local variable: 0x404050
Heap variable : 0x3a531010
Stack variable : 0x7ffc0486fb5c

Values:
Global variable = 100
Static variable = 200
Static local = 400
Heap variable = 500
Stack variable = 300
jella_pragathi@LAPTOP-RQQG555:~/OSSP$ nano session7_maps.c
jella_pragathi@LAPTOP-RQQG555:~/OSSP$ gcc -Wall -Wextra -g session7_maps.c -o session7_maps
jella_pragathi@LAPTOP-RQQG555:~/OSSP$ ./session7_maps
Memory Analysis Program
=====
PID: 589
Global address : 0x5dcdb6c73010
Heap address : 0x5dcdf6508010
Stack address : 0x7ffe737aa6fc

Process is running...
Press Ctrl+C to terminate.
^C
jella_pragathi@LAPTOP-RQQG555:~/OSSP$
```

```
jella_pragathi@LAPTOP-RQQG555:~$ cd ~/OSSP
cat /proc/589/maps
5dcdb6c6f000-5dcdb6c70000 r--p 00000000 08:30 26111 /home/jella_pragathi/OSSP/session7_maps
5dcdb6c70000-5dcdb6c71000 r-xp 00001000 08:30 26111 /home/jella_pragathi/OSSP/session7_maps
5dcdb6c71000-5dcdb6c72000 r--p 00002000 08:30 26111 /home/jella_pragathi/OSSP/session7_maps
5dcdb6c72000-5dcdb6c73000 r--p 00002000 08:30 26111 /home/jella_pragathi/OSSP/session7_maps
5dcdb6c73000-5dcdb6c74000 rw-p 00003000 08:30 26111 /home/jella_pragathi/OSSP/session7_maps
5dcdf6508000-5dcdf6529000 rw-p 00000000 00:00 0 [heap]
78f336800000-78f336828000 r--p 00000000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
78f336828000-78f3369c0000 r-xp 00028000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
78f3369c0000-78f336a0e000 r--p 001c0000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
78f336a0e000-78f336a12000 r--p 0020d000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
78f336a12000-78f336a14000 rw-p 00211000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
78f336a14000-78f336a21000 rw-p 00000000 00:00 0
78f336b54000-78f336b57000 rw-p 00000000 00:00 0
78f336b5d000-78f336b5f000 rw-p 00000000 00:00 0
78f336b5f000-78f336b63000 r--p 00000000 00:00 0
78f336b63000-78f336b65000 r--p 00000000 00:00 0
78f336b65000-78f336b67000 r-xp 00000000 00:00 0
78f336b67000-78f336b68000 r--p 00000000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
78f336b68000-78f336b97000 r-xp 00001000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
78f336b97000-78f336ba2000 r--p 00030000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
78f336ba2000-78f336ba4000 r--p 0003b000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
78f336ba4000-78f336ba5000 rw-p 0003d000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
78f336ba5000-78f336ba6000 rw-p 00000000 00:00 0
7ffe7378b000-7ffe737ac000 rw-p 00000000 00:00 0 [stack]
jella_pragathi@LAPTOP-RQQG555:~/OSSP$ cat /proc/589/maps | grep heap [heap]
5dcdf6508000-5dcdf6529000 rw-p 00000000 00:00 0
jella_pragathi@LAPTOP-RQQG555:~/OSSP$ cat /proc/589/maps | grep stack [stack]
7ffe7378b000-7ffe737ac000 rw-p 00000000 00:00 0
jella_pragathi@LAPTOP-RQQG555:~/OSSP$ cat /proc/589/maps | grep session7_maps
5dcdb6c6f000-5dcdb6c70000 r--p 00000000 08:30 26111 /home/jella_pragathi/OSSP/session7_maps
5dcdb6c70000-5dcdb6c71000 r-xp 00001000 08:30 26111 /home/jella_pragathi/OSSP/session7_maps
5dcdb6c71000-5dcdb6c72000 r--p 00002000 08:30 26111 /home/jella_pragathi/OSSP/session7_maps
5dcdb6c72000-5dcdb6c73000 r--p 00002000 08:30 26111 /home/jella_pragathi/OSSP/session7_maps
5dcdb6c73000-5dcdb6c74000 rw-p 00003000 08:30 26111 /home/jella_pragathi/OSSP/session7_maps
```

```
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ pmap 589
589: ./session7_maps
00005dcd6c6f000 4K r---- session7_maps
00005dcd6c70000 4K r-x-- session7_maps
00005dcd6c71000 4K r---- session7_maps
00005dcd6c72000 4K r---- session7_maps
00005dcd6c73000 4K rw--- session7_maps
00005dcd6f6508000 132K rw--- [ anon ]
000078f33680000 160K r---- libc.so.6
000078f336828000 1632K r-x-- libc.so.6
000078f3369c0000 312K r---- libc.so.6
000078f336a0e000 16K r---- libc.so.6
000078f336a12000 8K rw--- libc.so.6
000078f336a14000 52K rw--- [ anon ]
000078f336b54000 12K rw--- [ anon ]
000078f336b5d000 8K rw--- [ anon ]
000078f336b5f000 16K r---- [ anon ]
000078f336b63000 8K r---- [ anon ]
000078f336b65000 8K r-x-- [ anon ]
000078f336b67000 4K r---- ld-linux-x86-64.so.2
000078f336b68000 188K r-x-- ld-linux-x86-64.so.2
000078f336b97000 44K r---- ld-linux-x86-64.so.2
000078f336ba2000 8K r---- ld-linux-x86-64.so.2
000078f336ba4000 4K rw--- ld-linux-x86-64.so.2
000078f336ba5000 4K rw--- [ anon ]
00007ffe7378b000 132K rw--- [ stack ]
total 2768K
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ |
```